

Cooperation Without Adaptation: LLM Agents in a Collective-Risk Dilemma

Anonymous Author(s)

Abstract

Language-model agents are increasingly deployed to act on behalf of people in settings where individual decisions shape shared outcomes. We examine whether five production models adapt their contributions to changing risks and partner behaviour, or instead follow a relatively fixed strategy. In a collective-risk dilemma game (CRP), six players contribute to a shared pool over ten rounds; if the pool falls short of its target, a catastrophe may destroy their remaining resources. Four models contributed approximately an equal share at every positive risk level, and four of the five also contributed when risk was zero, even though contributing could not improve their payoff. Explicitly stating that only the agent’s own payoff mattered reduced zero-risk contributions for most models, while a more neutral prompt did not. Several models correctly identified which action offered the higher expected payoff at different risk levels, yet their contributions changed little. Against fixed partners, no model consistently chose the payoff-maximising response, and in mixed-model groups, a single agent’s shortfall in the final round could cause an otherwise successful group to fail. Payoff-based selection also depended on population composition: in a single six-agent group, it favoured the model that contributed least and reduced group success, whereas in larger, randomly mixed populations, it favoured an exact fair-share model at moderate and high risk. These findings show that cooperative self-play alone is not evidence of strategic adaptability. Consistent with a machine-behaviour approach, evaluations of agents in multi-agent systems should test how they respond to incentives, partner actions, and population composition.

Keywords

LLM agents, collective-risk social dilemma, cooperation, best response, empirical game-theoretic analysis, multiagent evaluation

ACM Reference Format:

Anonymous Author(s). 2027. Cooperation Without Adaptation: LLM Agents in a Collective-Risk Dilemma. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 10 pages.

1 Introduction

Language-model agents increasingly act on behalf of people in settings where multiple agents jointly shape a shared outcome, such as budget allocation, shared compute use, or negotiations over common resources [7, 19]. These systems are therefore not only engineering artifacts: their behaviour in interactive settings is

also an object of empirical study, as emphasized by the emerging field of machine behaviour [41]. In many such settings, avoiding collective loss requires agents to incur individual costs for the benefit of the group [34, 43, 45]. This raises a central question for multi-agent systems (MAS) research: do language-model agents condition their cooperative behaviour on the strategic environment, or do they rely on relatively fixed behavioural defaults? From a system-design and machine-behaviour perspective, this question has three components. First, is an agent’s cooperation sensitive to the magnitude of collective risk? Second, does it respond to the behaviour of other agents in the group? Third, under selection pressures that retain or replicate higher-earning agents, which behavioural tendencies are favoured?

Success in self-play, a common criterion in MAS benchmarks [24, 40], does not by itself answer these questions. When all positions in a group are occupied by copies of the same model, an agent that conditions its decisions on the stakes may generate the same observed pattern of play as an agent that follows a fixed rule. A machine-behaviour approach therefore calls for testing agents across systematically varied environments, rather than inferring their decision policies from a single setting. Success rates likewise obscure the extent to which a group falls short of the attainable optimum: a group may achieve the collective target while still dissipating a substantial share of its endowment. Moreover, a prompt that specifies both a group target and a fixed number of rounds may implicitly suggest an equal division of the burden, causing agents that simply reproduce this split to appear cooperative under conventional summary measures.

We study these questions using the collective-risk social dilemma game (CRG), a game introduced to study collective decision making for climate change mitigation [34]. Players contribute into a shared pool over several rounds, and if the pool misses a target, they all lose their savings with probability p (Figure 1a). For risk-neutral players, group welfare is maximised when no one contributes if $p < \frac{1}{2}$, and when the group meets the target exactly if $p > \frac{1}{2}$, for example through the *fair share*, defined as an equal split of the target. The game also contains two situations in which contributing cannot raise a player’s own payoff: when there is no risk, and once the outcome is settled. Neither risk attitude nor concern for the group can explain contribution in these situations, so they separate a default from a response to the stakes.

We introduce a game-theoretic evaluation that separates cooperative outcomes from incentive-sensitive behaviour by combining risk variation, dominated-action tests, exact best responses to fixed partners and selection in mixed populations, where each model is a strategy of an empirical game [56, 58]. We apply it to five production models, Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B and Grok 4.20, in four designs (Figure 1b), each compared with a benchmark computed from the game. We use standard equilibrium and finite-population tools, not a new solution concept. We call near-flat contribution across risk levels under a



This work is licensed under a Creative Commons Attribution International 4.0 License.

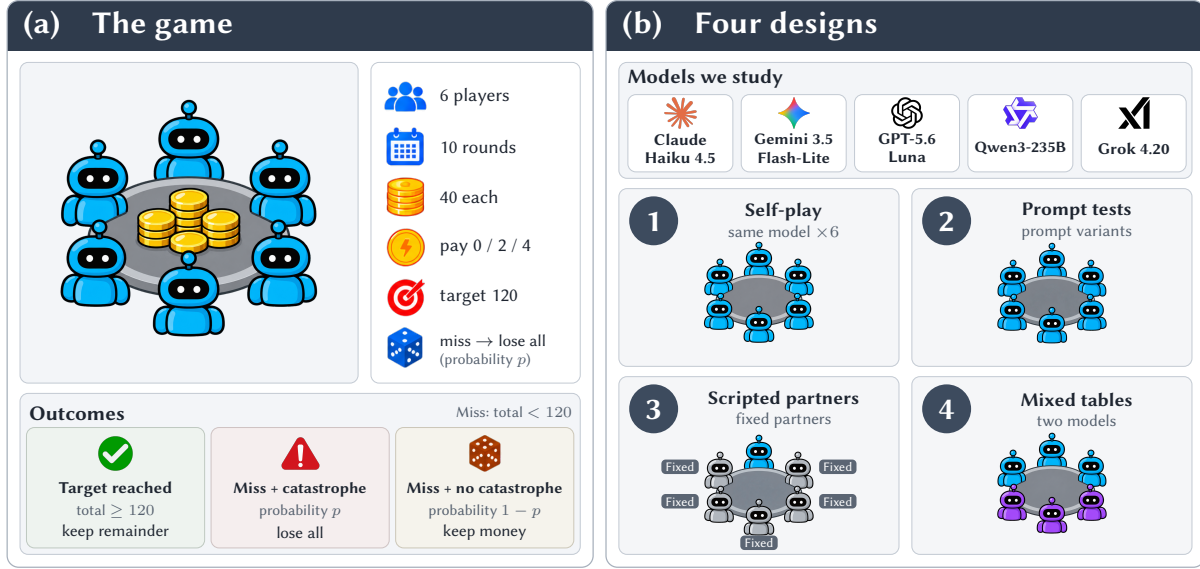


Figure 1: Overview. (a) The game (also referred to as a table or group, Section 3.1): six players hold 40 units each and have to decide to contribute either 0, 2 or 4 into a shared pool in each of ten rounds; if the pool misses the target of 120, a catastrophe destroys all their savings with probability p . (b) The five models and the four designs (Table 2): (1) self-play over the risk grid; (2) self-play under prompt and decoding controls, and with in-game value questions; (3) one model beside five scripted partners that follow fixed rules; (4) two models at one table, one of them in one to five of the six seats and the other in the rest. Selection (Section 3.3) uses the base-prompt payoffs of designs 1, 2 and 4.

fixed prompt a *default*; the term describes behaviour, not internal reasoning.

We report three sets of findings, one per question. (1) *Stakes*. Most models contribute roughly the fair share at every positive risk level, even when there is no risk. Removing normative wording does not change this pattern. Stating that only the agent’s own payoff matters reduces most zero-risk contributions, but only GPT-5.6 Luna shifts from contributing little at low risk to contributing the fair share at high risk. Asked in separate calls, several models rank the two choices correctly and change that ranking with the risk, but they keep contributing the same amount (Section 4.1). (2) *Partners*. Beside scripted partners who make every contribution a loss, no model plays the best response of contributing nothing, and two keep contributing after the outcome is already settled. In a mixed game (also referred to as a table or group), a single partner that stops contributing in the last round often causes a table of exact fair-share players to fail, while a table that overpays can absorb the shortfall. (Section 4.2). (3) *Selection*. Population size changes who is selected under payoff-based imitation: a population of one table favours the smallest contributor and lowers group success, whereas a larger randomly mixed population favours the exact fair-share player at moderate and high risk (Section 4.3).

2 Related work

Collective risk and selection. The collective-risk dilemma was introduced to study why groups fail to avoid a shared disaster, and human groups reach the target mostly when the risk is high [34]; inequality and communication change how often they succeed [51].

Threshold public goods games have many equilibria [37], contributions decline over the rounds of repeated public goods games [3], and the risk of collective failure can turn a defection trap into a coordination problem [33, 43]. In evolutionary simulations of the same game, constant fair sharers lose out to players who hold back early and compensate later [1]. Selection on payoff relative to a small group favours whoever does better than its neighbours [46], while selection between groups can favour cooperation [54]. Finite-population analyses price the incentives that sustain cooperation in this game [11] and show that maximising cooperation need not maximise welfare [21]. This literature predicts that behaviour should vary with risk, so the absence of such variation is informative; we use a similar game theoretical approach to analyse how different agent types perform when placed together in mixed groups/tables.

Language models as strategic players. Language models have been studied as economic subjects [8, 15, 32] and in repeated two-player games [2, 16, 36]. Models tend to be more cooperative than people [32], and an even split can reflect a focal point rather than fairness [10]. The weight of framing relative to game structure differs by model [31], and models can hold a correct belief or plan and act against it [12, 47]. FAIRGAME runs language-model agents through standard games and finds outcomes that depend on model, language and persona [5]; later work extends it to a payoff-scaled prisoner’s dilemma and a multi-player public goods game [26].

Language models in social dilemmas. Societies of agents over-use a common resource [40], contributions to a public good can stay the

same even when it returns nothing [24], and reasoning models free-ride [18]. Agents stand in for people in a collective-risk game [49] and for developers, regulators and users in an evolutionary model of AI regulation [4]. A study of 35 open models includes a collective-risk game at one risk level and observes their decisions far from the group optimum [28]. In a binary collective-risk game with strategies written by three models, payoff-biased imitation drives larger groups to selfish, low-welfare outcomes [59], and prompts that evolve under individual selection collapse into defection while group selection keeps them prosocial [6]. A recent study audits rule recall, state tracking and payoff calculation before interpreting play in multi-player races to develop AI [39]. It compares play with an evolutionary benchmark and human data, and finds that correct recall can coexist with weak expected-payoff calculation. We differ in three ways: we vary the risk from zero to certainty and add two tests in which contributing cannot raise the contributor’s cash; we compute the exact best response to fixed partners; and we select among deployed models with payoffs from their real mixed tables, finding that at moderate and high risk the favoured model changes once the population exceeds one table.

Multiagent evaluation. Multiagent reinforcement learning studies cooperation in sequential social dilemmas [25, 30], and ad hoc teamwork and cooperative AI ask for agents that work with partners unlike themselves [9, 29, 50]. Empirical game-theoretic analysis treats trained agents as strategies of a meta-game [56, 58], α -Rank ranks them by evolutionary dynamics [35], and evolutionary game theory is used to study emergent behaviour in multiagent systems [20]. Language models show distinct strategic fingerprints in tournaments with classic strategies [38], their cooperation evolves differently across base models [57], and cooperation sustained by repetition weakens when co-players differ [53]. In the collective-risk game, people who delegate to artificial agents succeed more often [14], and in theory adaptive players compensate for the effort of artificial agents [52]. We add a multiplayer dilemma with exact best responses to fixed partners, and measure this compensation in real models.

3 Game and evaluation design

3.1 Collective-risk game and benchmarks

Six players each start with $e = 40$ units, and the game lasts $R = 10$ rounds. In each round every player privately chooses a contribution $a_i^t \in \{0, 2, 4\}$ to a shared *pool*, and after each round all players see what every seat has contributed. Let $x_i = \sum_t a_i^t$ be a player’s total and $X = \sum_i x_i$ the pool. The target is $T = 120$. A player keeps what it did not contribute, but if the pool misses the target, a catastrophe destroys all savings with probability p (Table 1 lists the terms used):

$$u_i = \begin{cases} e - x_i & \text{if } X \geq T, \\ 0 \text{ with probability } p, \text{ else } e - x_i & \text{if } X < T. \end{cases} \quad (1)$$

Proposition 1. *Consider the expected-payoff formulation of the game in which each player picks a total $x_i \in \{0, 2, \dots, 40\}$, and let $p < 1$. A profile is a Nash equilibrium if and only if $x = 0$, or $X = T$ and $x_i \leq p \times e$ for every i . A target-reaching equilibrium exists if and only if $p \geq p^* = T/(6e) = \frac{1}{2}$. The welfare-optimal outcome is $x = 0$ for $p < p^*$ and $X = T$ for $p > p^*$, and both are*

Table 1: Symbols and terms.

Term	Meaning
Seat; table	One player position; the six seats of one game
Fair share	The target split equally: 20 per seat, or 2 per round
Exact split	Contributing 2 in every one of the ten rounds
Pivot $p^* = \frac{1}{2}$	Risk above which reaching the target is optimal
$\text{opt}(p)$	Welfare-optimal expected payoff per seat
Dominated	Lowers the seat’s own payoff whatever others do
Settled	Further contributions cannot affect the outcome: the pool has already reached T or can no longer do so.

optimal at $p = p^$, so the welfare-optimal mean payoff per player is $\text{opt}(p) = \max\{(1 - p)e, e - T/6\}$; this welfare statement also holds at $p = 1$.*

Proof sketch. If $X < T$, any player with $x_i > 0$ gains by contributing nothing, so the only equilibrium that misses the target is $x = 0$. If $X > T$, then $X \geq T + 2$ because totals are even, so a player can contribute 2 less and still reach the target. If $X = T$, the best deviation is to contribute nothing, worth $(1 - p)e$ against $e - x_i$, which gives $x_i \leq pe$; summing over players gives $p \geq T/(6e)$, and conversely $x_i = 20$ for every i is an equilibrium when $p \geq \frac{1}{2}$. Welfare is $6e - X$ when the target is met and $(1 - p)(6e - X)$ otherwise, so it peaks at $X = T$ or $X = 0$. The ten-round game, in which strategies can react to the history, has the same totals in pure-strategy Nash equilibrium; the Supplementary Material gives the full proof (Proposition S1, Section S2.3). We claim nothing about subgame perfection, and $\text{opt}(p)$ is a welfare bound that does not depend on the equilibrium concept. $\square$

What we compare play with. Proposition 1 gives a *symmetric welfare benchmark*: keep everything below the pivot and contribute the fair share above it. Asymmetric allocations reaching exactly T attain the same group welfare. This benchmark is not a unique equilibrium or a prediction of individual play: contributing nothing is also an equilibrium at every risk level. Contributing is dominated in own payoff at $p = 0$ and, for $p < 1$, after the outcome is settled. These tests do not require risk neutrality or beliefs about the partners. They do assume that larger own payoff is preferred; non-payoff goals are not ruled out.

Proposition 2. *For two seats i and j in the same game, the difference in their expected payoffs over the lottery is $c(x_j - x_i)$, with $c = 1$ if the target was reached and $c = 1 - p$ otherwise.*

Both seats face the same pool and the same lottery, so only their own contributions differ. Hence, for $p < 1$, whoever contributes less at a table earns more, whatever the group achieves.

3.2 Models, protocol and measures

We test the five models of Section 1: one production model per provider, chosen to reflect the kind of model plausibly used for deploying many agents at scale. Qwen3-235B is the instruction-tuned build without a thinking mode, and Grok 4.20 is its non-reasoning variant.

Prompts. In each round a seat receives a prompt with the rules, the full history of every seat’s contributions, and a request for

Table 2: Designs. A cell is one condition at a risk level: one model in self-play, one model with a scripted partner, or one pair of models with a split of seats. Each cell holds ten games.

Design	Risk levels p	Games
Self-play grid	0, 0.1, ..., 1	550
Rerun, with questions	0.1, 0.5, 0.9	150
No hint	0.1, 0.5, 0.9	150
Neutral wording	0, 0.1, 0.9	150
Neutral wording, goal stated	0, 0.1, 0.9	150
Two rewordings	0.1, 0.9	200
Temperature zero	0.1, 0.9	100
One model, five scripts (4 kinds)	0.1, 0.3, ..., 0.9	1,000
Two models (10 pairs, 1 to 5 seats)	0.1, 0.5, 0.9	1,500
All		3,950

making a contribution; the Supplementary Material provides every prompt word for word (Section S4). The base prompt calls the pool a “climate account”, says that the group “must” reach the target, calls the loss a “disaster” and names the game as a collective-risk social dilemma. It never states the player’s goal, and it hints at the equal split twice. The prompt controls in Section 3.3 change exactly these parts.

Protocol. The running pool is not shown, so any seat that needs it must compute it from the history. The prompt is in English and no seat is assigned a persona. We parse the move from the reply’s final CONTRIBUTION: line.

Measures. We report *expected* payoffs, the average a seat would earn if the catastrophe lottery were run many times on the same contribution, and the share of $\text{opt}(p)$ that is lost. Proposition 1, $\text{opt}(p)$ and the selection fitness below all use expected payoffs, which remove the lottery noise without bias; agents never see the draw, and the Supplementary Material lists the realised draws (Table S6). The unit of analysis is the game, since seats in one game share a history and a lottery. Intervals are 95% percentile bootstraps over games and tests are permutation tests over games; Supplementary Material gives every value behind the statements below (Sections S6–S12).

3.3 Evaluation designs

Table 2 lists the designs; every model plays the same number of games in each.

Self-play and prompt controls. Six copies of one model play the base prompt at 11 risk levels $p = 0, 0.1, \dots, 1$, and again under prompt and decoding controls at fewer risk levels. *No hint* deletes the two equal-split phrases. *Neutral wording* replaces the normative and naming phrases with plain ones, such as “a shared account”. *Goal stated* keeps the neutral wording and adds an opening that says the player is in a paid experiment where “the only thing that matters to you is your own final cash payoff”. Two *rewordings* change the phrasing but keep every rule and the hint, one condition sets the temperature to zero, and a *rerun* repeats the base prompt with value questions.

Value questions. In the rerun, separate calls whose answers never enter the game ask the first seat at the start, middle and end of the game which of two whole-game strategies gives the higher average payoff: contributing 2 in every round and reaching the target, or contributing nothing and missing it. The correct answer switches at p^* (Supplementary Material, Section S8).

Fixed partners. One model plays with five scripted partners of the same type. They always contribute 0, always contribute 2, always contribute 4, or copy the group, which means contributing 2 in the first round and then the allowed amount nearest to the average of what the other seats contributed in the previous round. The model is not informed that its partners are scripted, but their moves are included in the history. We compute the *best response* to each type, i.e. the play that earns the most, by enumerating every possible sequence of the model’s own ten moves. With partners who always contribute 0, the target cannot be reached; with partners who always contribute 4, the target is reached without the model. The best response to both is to contribute nothing, and once the history settles the outcome, contributing cannot improve the outcome and only reduces payoff. Against fair-share partners, the best response is to contribute nothing below p^* and exactly the fair share above it; partners who copy the group never deviate from 2, thus resulting in the same best response.

Mixed tables. To mix defaults at a table, we seat every pair of models together, with one to five seats for the first model and the rest for the second, at low, moderate and high risk. Each model’s seats form one block, fixed within a cell and alternating across cells, so every model occupies every seat equally often overall.

Selection among models. We finally examine which model prevails under payoff-based retention or imitation, and how this outcome depends on population size. We treat each LLM as a strategy in an empirical game, i.e. using experimentally estimated payoffs from model interactions to define the game [56, 58]. We write $\pi_A(k)$ for the mean expected payoff of an A seat at a table with k seats of A and $6 - k$ of B , from mixed tables for $1 \leq k \leq 5$ and from self-play (grid plus rerun) for $k = 6$. A population of N agents consists of i agents of type A and $N - i$ of type B , and each agent’s five tablemates are drawn at random from the other $N - 1$, as in e.g. [22, 43, 44]. The fitness of each type is its payoff averaged over the tables/interactions it participates in:

$$f_A(i) = \sum_{k=1}^6 \mathcal{H}(k-1; N-1, i-1) \pi_A(k), \quad (2)$$

$$f_B(i) = \sum_{k=0}^5 \mathcal{H}(k; N-1, i) \pi_B(6-k),$$

where $\mathcal{H}(k; N-1, m)$ is the hypergeometric chance that k of five tablemates come from the m other agents of type A . Under Fermi imitation [22, 55], an randomly selected agent copies a random other agent’s strategy with probability $1/(1 + e^{-\beta \Delta f})$, where Δf is the other’s fitness minus its own and $\beta = 1$ per payoff unit. We assume rare mutation [17, 22], which has been shown to provide a strong approximation beyond the strict rare-mutation regime [23, 42, 48, 60]: each newcomer either fixes or disappears before the next arrives. Pairwise fixation probabilities then define a Markov chain over the five homogeneous model populations. Its *stationary mass* gives each model’s long-run weight, and EGTtools [13] was used to reproduce the fixation probabilities and stationary distributions

of Figure 5. With $N = 6$ the whole population is one table; with $N = 30$, fitness averages over random, overlapping tables. Changing N changes both table compositions and fixation dynamics, not only whom an agent is compared with. We evaluate this finite-population process, not a scoring-only intervention.

4 Results

4.1 Stakes: contribution does not scale with risk

4.1.1 Self-play across risk. Four of the five models contribute approximately the fair share at every positive risk level (Figure 2a). Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B contribute almost the same amount below the pivot as above it (a step of at most 0.2 units), whereas the symmetric welfare benchmark jumps from contributing nothing to the fair share. In the original experiment with the same rules, human groups reached the target in half of their games at $p^* = 0.9$ and in none at $p^* = 0.1$ [34]. Strong risk aversion could explain a flat contribution profile above zero risk: because catastrophe leaves players with nothing, a player with relative risk aversion near one may prefer a certain fair-share contribution even at low risk. Gemini 3.5 Flash-Lite provides the clearest example. Almost every seat plays the exact split at every risk level (Table 3), so this near-fixed rule attains the welfare optimum above the pivot but not below it.

Paying when there is nothing to protect. Four of the five models also contribute at $p = 0$, about as much as at any other risk level (Table 3, first column). Contributing there only lowers a player’s own payoff and helps nobody, so neither risk attitude nor concern for others’ payoff can explain this contribution. GPT-5.6 Luna is the exception. It contributes nothing at $p = 0$ and its usual share at every positive risk, so it distinguishes zero risk from positive risk but does not respond to how high the risk is.

Three ways to lose. The models lose payoff in three different ways (Table 3, Figure 2c). Gemini 3.5 Flash-Lite, Claude Haiku 4.5 and GPT-5.6 Luna lose mostly at low risk, where they contribute for a target that is not worth reaching. Grok 4.20 overpays: its groups reach the target well before the last round, and its seats keep contributing into a pool that is already safe. Qwen3-235B fails at the end. It contributes 2 round after round, then almost all of its seats contribute nothing in the last round (Figure 2b). Its groups therefore almost never reach the target, although every one of them would have succeeded had each seat simply repeated its previous move. Success rates hide all of this. Gemini 3.5 Flash-Lite and Grok 4.20 both reach the target in every game, yet Grok 4.20 keeps far less of its endowment. For $0.5 < p < 1$, Qwen3-235B and Grok 4.20 groups earn less in most games than in the worst equilibrium of Proposition 1, in which nobody contributes. In the last round, Qwen3-235B skips as often when its contribution is needed as when the others’ fair shares already suffice, so its skip follows the round number rather than the pool; human contributions similarly decline over the rounds of repeated public goods games [3]. GPT-5.6 Luna skips mostly when its contribution is not needed, and Claude Haiku 4.5 stops mostly once the target is reached (Supplementary Material, Section S12.1).

4.1.2 Prompt and objective controls.

Table 3: Self-play over the risk grid. Pays at $p=0$: mean seat total where contributing is dominated. Exact split: seats contributing 2 in all rounds. Target: games reaching it. Payoff: expected payoff per seat; the optimum averages 25.5. Lost: share of $\text{opt}(p)$ lost. Main loss: contributing below p^* , contributing past 120, or missing the target at or above p^* (in the last round).

Model	Pays at $p=0$	Exact split (%)	Target (%)	Payoff	Lost Main (%) loss
Claude Haiku 4.5	20.9	17	98	18.9	26 below p^*
Gemini 3.5 Flash-Lite	20.0	99.7	100	20.0	21 below p^*
GPT-5.6 Luna	0.0	72	70	19.2	25 below p^*
Qwen3-235B	18.0	4	2	10.9	57 missed
Grok 4.20	37.4	2	100	5.0	80 past 120

The default effect persists under rewording, but the resulting failures do not. Deleting the two equal-split phrases does not reduce any model’s contribution (Table 4), indicating that the hint was not the source of contributions. Instead, it bounded contributions for some models while enabling others contribute exactly 2 per round. Gemini 3.5 Flash-Lite still plays the exact split, which it can rebuild from the target, while GPT-5.6 Luna keeps its average contribution but plays the exact split less often. Qwen3-235B and Grok 4.20 contribute more without the hint; Qwen3-235B now reaches the target, but it earns less because it overpays. The two rewordings and a temperature of zero leave the contributions of the four models other than Grok 4.20 almost unchanged. The failures are less stable than the averages. Under the second rewording Grok 4.20 contributes about the fair share, and how often Qwen3-235B drops out in the last round varies widely across prompts. Grok 4.20’s rise with risk on the grid recurs at temperature zero but not in the rerun or the second rewording, so we do not interpret it as a stable response.

Stating the goal, not removing norms, reduces zero-risk contributions. The normative words are not the main source of the default. With neutral wording, most models still contribute at $p = 0$. At low and high risk the four models other than Grok 4.20 contribute on average what they contributed before, while Grok 4.20 contributes even more and GPT-5.6 Luna plays the exact split less often (Table 4). Adding the stated goal is what changes play at zero risk: Claude Haiku 4.5 and Gemini 3.5 Flash-Lite then contribute almost nothing, Qwen3-235B contributes far less than before, and Grok 4.20 barely changes. The goal still does not make most models weigh the risk. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite now separate zero from positive risk but already contribute the fair share at low risk, a pattern that strong risk aversion could also produce. Only GPT-5.6 Luna changes from contributing little at low risk to the fair share at high risk. Qwen3-235B contributes slightly more at high than at low risk but less than before at every risk level, and never reaches the target; Grok 4.20 still overpays.

For most models, behaviour at zero risk changes depending on whether the prompt clearly states the goal. When risk above zero, models tend to contribute similar amounts across risk levels, and this pattern remains under different wordings. The partner and selection results below use the base prompt.

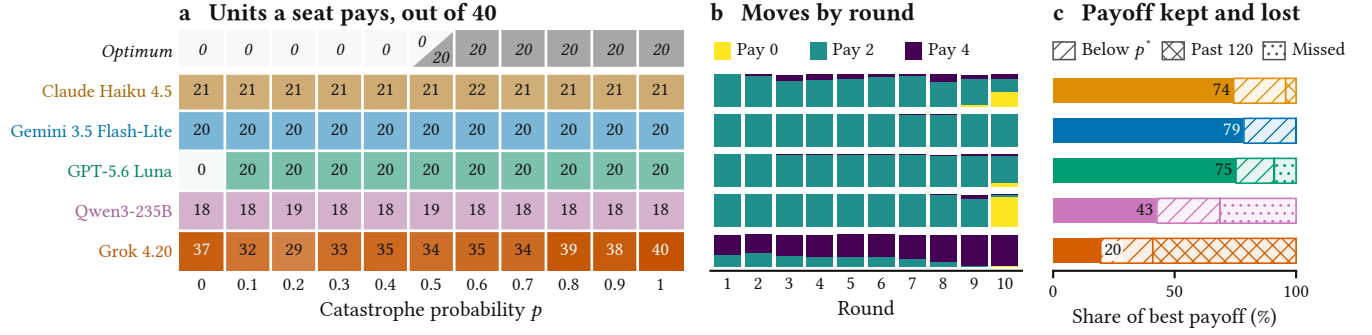


Figure 2: Self-play on the risk grid, 110 games per model. (a) Mean units a seat contributes out of 40, ten games per cell; the top row is the optimum of Proposition 1, where both 0 and 20 are optimal at $p = 0.5$. (b) Share of seats contributing 0, 2 or 4 in each round, over the games with $p > 0$. (c) Expected payoff kept as a share of $\text{opt}(p)$, and the loss split into contributing toward the target below p^* , contributing past 120, and missing the target at or above p^* .

Table 4: Mean seat total under prompt and decoding changes. Top: pooled over $p = 0.1$ and 0.9 (20 games each). *Rerun* repeats the base prompt with in-game questions; *Para. 1, 2* are the rewordings; *Wording* is the neutral wording and *Goal* adds the stated goal. Bold: a shift from *Base* of at least 2 units with permutation $P < 0.01$; every bold shift also has $P < 0.001$, below the Bonferroni threshold for the 35 comparisons. Bottom: single risk levels (10 games each).

Condition	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
Base	20.9	20.0	19.9	18.4	35.1
Rerun	21.3	20.0	19.9	18.5	35.6
No hint	21.7	20.0	19.9	34.7	40.0
Para. 1	21.3	20.0	19.9	19.4	31.8
Para. 2	20.9	20.0	19.8	18.0	20.6
Temp. 0	21.2	20.0	20.0	18.3	35.1
Wording	21.0	20.0	19.4	18.7	39.7
Goal	20.0	20.1	13.9	8.2	37.7
Base $p=0$	20.9	20.0	0.0	18.0	37.4
Wording $p=0$	18.0	19.9	0.4	18.1	39.8
Goal $p=0$	0.3	1.4	0.2	4.8	36.7
Goal $p=0.1$	19.8	20.2	7.7	6.4	36.7
Goal $p=0.9$	20.2	20.0	20.1	10.1	38.7

Correct answers, unchanged play. Asked the value question in separate calls, Claude Haiku 4.5, Gemini 3.5 Flash-Lite and GPT-5.6 Luna mostly answer correctly and change their answer with the risk, yet the questioned seats contribute almost the same at low and high risk; Qwen3-235B and Grok 4.20 mostly answer that keeping pays more. This is a gap between answers and play, not evidence about what an agent computes; the Supplementary Material gives every answer and the answers against play (Section S8.6). A strongly risk-averse agent could give these answers and still contribute; contribution at $p = 0$ and after settlement provides the sharper test.

4.2 Partners: simple responses and fragile tables

4.2.1 Fixed partners.

Table 5: Units contributed per game before and after the outcome is settled, alongside five partners who always contribute 0 or always contribute 4 (50 games per entry). Given these partners, every unit contributed is wasted; units contributed after settlement are wasted even for an agent unaware that its partners are scripted.

Model	Always 0		Always 4	
	Before	After	Before	After
Claude Haiku 4.5	7.7	0.1	14.2	1.1
Gemini 3.5 Flash-Lite	13.6	0.2	10.9	0.0
GPT-5.6 Luna	8.5	0.0	4.5	0.0
Qwen3-235B	17.8	12.1	12.3	8.1
Grok 4.20	19.6	11.1	16.7	16.2

No model keeps all its endowment when that is the best response. The model contributes nothing only when paired with partners who always contribute either 0 or 4. Of the 1,000 scripted games, only 186 are played exactly optimally, and in every case the optimum is the fair share. In other words, the models reach the optimum only when it coincides with the default contribution. Since an agent unaware that its partners are scripted may reasonably contribute early to learn their behaviour, the cleaner evidence comes from contributions made after the outcome has settled (Table 5).

Partners matter, risk does not. We split the variation in what a model contributes across games with three fixed-amount partners into partner, risk and interaction parts. For Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B, partner type explains most of the variation (69% to 92%), while risk and its interaction with partner type explain at most 5%. For an exact best responder in the same games, risk would explain half of the variation. Beside fair-share partners, where the best response jumps from nothing to the fair share at p^* , no model’s total contribution changes by more than 2 between the low- and high-risk conditions. Grok 4.20’s total follows neither pattern: it is driven largely by its opening move, which is its largest contribution in around half of its games.

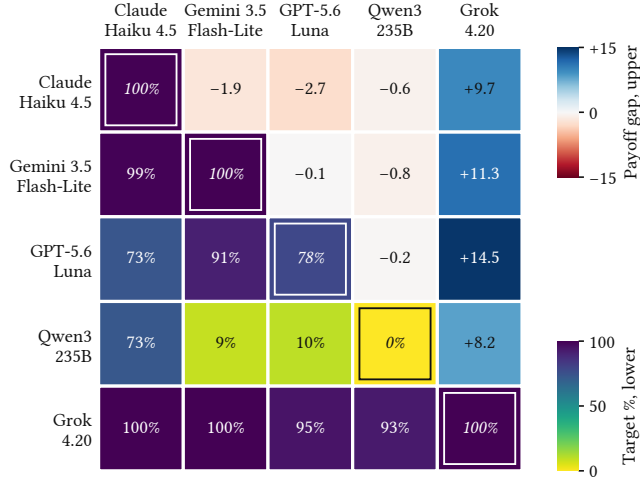


Figure 3: Two models play at each table, with 150 games per pair, pooled across seat assignments and risk levels. Above diagonal: expected payoff of row model’s seats minus column model’s, in the same games. On and below the diagonal: proportions of games reaching the target. Boxes mark self-play (60 games: grid games plus a rerun at same risks).

Responses to partners take simple forms. The models adjust their contributions in response to their partners, but not in the way a best responder would (Table 5; Figures S4 and S5 of the Supplementary Material show every round). Gemini 3.5 Flash-Lite contributes its usual amount, while both it and Claude Haiku 4.5 stop contributing once the history indicates that the outcome is settled. With partners who always contribute 4, GPT-5.6 Luna contributes little even the outcome is settled, remaining close to a best response as it learns its partners’ behaviour. Qwen3-235B contributes more, not less, when interacting with partners who contribute nothing. Both Qwen3-235B and Grok 4.20 also continue making large contributions after the outcome is settled – a behaviour that no belief about their partners can justify. Against five fair-share partners, Qwen3-235B contributes 2 for nine rounds and then nothing, so the table misses the target in almost every game by that single payment; Gemini 3.5 Flash-Lite in the same seat reaches it every time.

4.2.2 Mixed tables. At a shared table every model earns more than Grok 4.20, while the payoff differences among the other four models are small (Figure 3, above the diagonal). This is consistent with Proposition 2: Grok 4.20 contributes the most, and at a two-model table the higher contributor earns less. Group success depends on the pair (Figure 3, below the diagonal). Tables that include Grok 4.20 reach the target in the great majority of games, while tables that mix Qwen3-235B with Gemini 3.5 Flash-Lite or GPT-5.6 Luna rarely do.

An exact fair-share target leaves no margin for error. One partner who stops contributing in the final round can cause a table of fair-share players to fail (Figure 4a). When the pool is one round of fair shares short before the last round, Qwen3-235B contributes nothing in 92% of cases and Gemini 3.5 Flash-Lite in 0.1%. Five Gemini 3.5

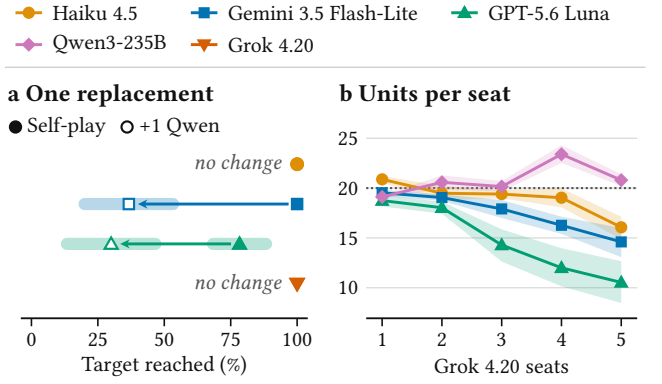


Figure 4: Mixed groups, pooled over $p = 0.1, 0.5, 0.9$. (a) Self-play (filled; 60 games, grid plus rerun) versus one Qwen3-235B replacement (open; 30); arrows run from the first to the second. (b) Mean contribution per focal seat beside one to five Grok 4.20 seats (30 games each); dotted: fair share. Bars and bands: 95% game-bootstrap intervals, not stratified by risk. Lines join measured means.

Flash-Lite seats with one Qwen3-235B seat succeed in only 11 of 30 games, although Gemini 3.5 Flash-Lite alone never misses in our sample; tables with GPT-5.6 Luna also succeed less often. Tables of Claude Haiku 4.5 or Grok 4.20, which contribute more, absorb the same partner. Replacing *all* Qwen3-235B last-round zeros by 2, holding other recorded actions fixed, would rescue 92% of the 323 failed mixed tables containing it. This is an arithmetic counterfactual, not a tested intervention. Conversely, one Grok 4.20 seat helps five Qwen3-235B seats succeed in most games, contributing for the rescue. These comparisons depend on the base prompt and the fixed seat blocks.

Agents free-ride on generous partners. Most agents contribute less as the number of Grok 4.20 seats increases from one to five (Figure 4b). Alongside five Grok 4.20 seats, GPT-5.6 Luna contributes 10.5 units, and Gemini 3.5 Flash-Lite and Claude Haiku 4.5 also fall below the fair share; these two stop once the pool reaches the target, while GPT-5.6 Luna contributes less from the start. Qwen3-235B does not show this decline. Within a cell, Grok 4.20’s higher opening contributions are associated with lower total contributions by Gemini 3.5 Flash-Lite and Claude Haiku 4.5, mainly through earlier target completion. GPT-5.6 Luna also contributes less while the pool is short, consistent with the compensation predicted alongside artificial agents [52]. Openings are not randomised and partners can react to a deviation, so this association identifies neither a causal response nor an exact best response.

4.3 Selection: population-size dependent

In a one-table population, selection favours the model that contributes least. Proposition 2 explains why: within a game, contributing less leads to a higher individual payoff relative to the other players. In our data, Qwen3-235B contributes least relative to its tablemates, so selection gives it the greatest weight (Figure 5a). But this individual advantage comes at a cost to the group: tables reach

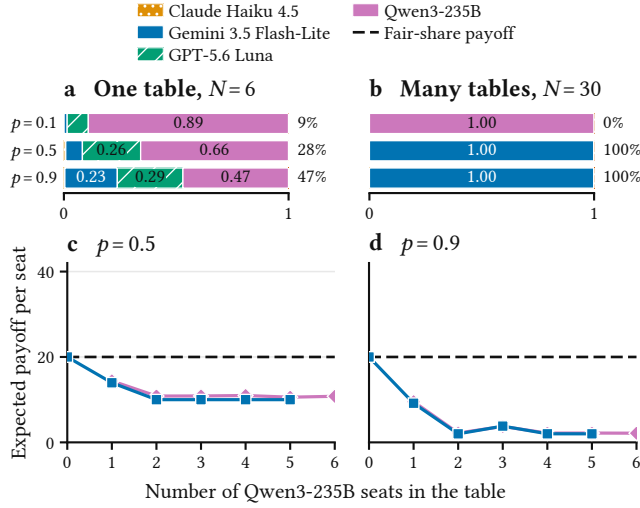


Figure 5: Rare-mutation Fermi selection ($\beta = 1$). (a, b) Stationary mass in a population of one table ($N = 6$) or a larger randomly mixed population ($N = 30$); right of each bar, the self-play target rate weighted by mass. (c, d) Expected payoff of a Qwen3-235B or Gemini 3.5 Flash-Lite seat by the number of Qwen3-235B seats at its table, at $p = 0.5$ and $p = 0.9$; dashed: fair-share payoff. Invasion diagrams are in the Supplementary Material (Figure S7).

the target less often when models are selected this way than when a model is chosen uniformly at random. At $p = 0.9$, for example, the target is reached in 47% of selected games, compared with 76% of randomly selected games. This illustrates the known result that selection based on payoffs relative to a small group favours whoever does better than its neighbours, even when this comes at a cost to the group [46]. Here, we observe this dynamic in games played by deployed language models.

In larger populations, selection favours Gemini 3.5 Flash-Lite at moderate and high risk (Figure 5b). Its self-play attains the welfare optimum there, but neither optimality nor Nash equilibrium implies that imitation will select it. The ranking comes from the estimated payoffs at every composition along the fixation path, where a lower contributor can earn more than its tablemates even while lowering its own payoff (Figure 5c, d). At low risk both population sizes favour Qwen3-235B, whose outcome remains far below the optimum. The switch to Gemini 3.5 Flash-Lite occurs already at seven or eight agents, consistent with finite-population theory [46]. Each winner is stable under bootstrap resampling, stronger selection, larger N and grid-only payoffs, but not under weak selection or when a candidate is removed (Supplementary Material, Sections S11.4 and S11.6).

5 Conclusion and discussion

Our results show that cooperative-looking self-play is not sufficient evidence that an agent responds appropriately to incentives or adapts to its partners. Across the five models, contributions often remain close to the fair share as risk changes, including when

contributing is payoff-dominated; some models also contribute after their actions can no longer affect the outcome. In mixed tables, one late shortfall can make an exact fair-share target fail. Selection outcomes depend on population structure: in a one-table population, selection can favour the lowest contributor at the group’s expense, while larger randomly mixed populations favour the fair-share model at moderate and high risk.

These findings matter for how we understand machine behaviour. A cooperative action need not reflect a stable preference for cooperation or an understanding of its consequences: it may instead be a default response to the prompt or a learned focal-point strategy. This empirical approach to auditing agent behaviour—treating models as subjects for observation in varied, interactive environments—aligns with the call to systematically study machine behaviour [41]. Conversely, correctly identifying the payoff-maximising action when asked does not ensure that a model will choose it in play. We therefore distinguish what models can state from how they behave, and caution against treating self-play success as evidence of strategic competence, social preference, or reliable adaptation. Evaluations should vary incentives and partner behaviour, test mixed populations, and compare individual payoffs with group welfare. They should also report the welfare benchmark, identify dominated actions, and measure contributions after the outcome is settled. This can reveal wasted resources and cases where apparent cooperation does not improve welfare [21].

The mechanisms behind these behaviours remain uncertain: our data show choices, not reasoning. One possibility is that an equal split acts as a focal point that models reconstruct even without an explicit hint [10]. Another is that questions prompt calculation while requests to act elicit a behavioural default [12, 47]. Post-training may also encourage models to contribute their share unless instructed otherwise, consistent with findings that models are more cooperative than people [32] and that cooperation increases with the stakes in repeated prisoner’s dilemmas, contrary to evolutionary predictions [27]. Stronger reasoning is not necessarily a remedy: reasoning models can free-ride more in public-goods games [18]. These accounts are hypotheses, not explanations established by our experiments.

Threats, limitations and ethics. The goal sentence is an instruction tested in one phrasing and in self-play only, and without a group-payoff goal we cannot separate objective clarity from the direction of the instruction. The running pool is not shown, so arithmetic errors may explain some contribution after settlement, and the value questions omit $p = 0$. We test five models, one per provider, in English and one game; claims concern this panel, not language models in general. Ten games per cell limit power, and a zero-width bootstrap interval in a constant sample does not prove deterministic behaviour. Rewordings cover a small prompt space, welfare losses assume risk neutrality, the scripted, mixed and selection designs use the base prompt, fixed seat blocks confound composition with position, and selection assumes rare entry. We will release prompts, configurations, seeds and code; they reproduce the design, not the model transcripts. The study involves no human subjects or personal data, and we report failure modes so designers can test for them before trusting agents with shared resources.

References

- [1] Maria Abou Chakra and Arne Traulsen. 2012. Evolutionary dynamics of strategic behavior in a collective-risk dilemma. *PLoS Computational Biology* 8, 8 (2012), e1002652. doi:10.1371/journal.pcbi.1002652
- [2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (2025), 1380–1390. doi:10.1038/s41562-025-02172-y
- [3] James Andreoni. 1988. Why free ride? Strategies and learning in public goods experiments. *Journal of Public Economics* 37, 3 (1988), 291–304. doi:10.1016/0047-2727(88)90043-6
- [4] Alessio Buscemi, Daniele Proverbio, Paolo Bova, Nataliya Balabanova, Adeela Bashir, Theodor Cîmpeanu, Henrique Correia da Fonseca, Manh Hong Duong, Elias Fernández Domingos, Antonio M. Fernandes, Marcus Krellner, Ndidi Bianca Ogbo, Simon T. Powers, Fernando P. Santos, Zia Ush Shamszaman, Zhao Song, Alessandro Di Stefano, and The Anh Han. 2025. Do LLMs trust AI regulation? Emerging behaviour of game-theoretic LLM agents. arXiv:2504.08640 doi:10.48550/arXiv.2504.08640
- [5] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. 2025. FAIRGAME: A Framework for AI Agents Bias Recognition Using Game Theory. In *ECAI 2025 – 28th European Conference on Artificial Intelligence (Frontiers in Artificial Intelligence and Applications, Vol. 413)*. IOS Press, Bologna, Italy, 4097–4104. doi:10.3233/FAIA251300
- [6] Luis Celiktemel, Edward Eichhorn, Levin Brinkmann, Robin Schimmelpfennig, Aron Vallinder, Yaomin Jiang, Edward Hughes, and Iyad Rahwan. 2026. Group Selection Promotes Prosocial Prompts in Populations of LLM Agents. arXiv:2606.23343 doi:10.48550/arXiv.2606.23343
- [7] Alan Chan, Carson Ezell, Max Kaufmann, Kevin Wei, Lewis Hammond, Herbie Bradley, Emma Bluemke, Nitarshan Rajkumar, David Krueger, Noam Kolt, Lennart Heim, and Markus Anderljung. 2024. Visibility into AI Agents. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, Rio de Janeiro, Brazil, 958–973. doi:10.1145/3630106.3658948
- [8] Yiting Chen, Tracy Xiao Liu, You Shan, and Songfa Zhong. 2023. The emergence of economic rationality of GPT. *Proceedings of the National Academy of Sciences* 120, 51 (2023), e2316205120. doi:10.1073/pnas.2316205120
- [9] Allan Dafoe, Edward Hughes, Yoram Bachrach, Tantum Collins, Kevin R. McKee, Joel Z. Leibo, Kate Larson, and Thore Graepel. 2020. Open Problems in Cooperative AI. arXiv:2012.08630 doi:10.48550/arXiv.2012.08630
- [10] Eleonore Dodivers and Ismael Rafai. 2025. Uncovering the fairness of AI: exploring focal point, inequality aversion, and altruism in ChatGPT’s dictator game decisions. *Economics Bulletin* 45, 4 (2025), 1818–1825. https://www.accessecon.com/pubs/EB/default.aspx?PaperID=EB-25-00292&topic=Abstract
- [11] M. H. Duong, C. M. Durbac, and T. A. Han. 2026. Cost of Institutional Incentives for Promoting Cooperation in 2 × 2 Games and Collective Risk Games. *Dynamic Games and Applications*. Online first, 7 February 2026. doi:10.1007/s13235-025-00696-6
- [12] Caoyun Fan, Jindou Chen, Yaohui Jin, and Hao He. 2024. Can Large Language Models Serve as Rational Players in Game Theory? A Systematic Analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. AAAI Press, Washington, DC, USA, 17960–17967. doi:10.1609/aaai.v38i16.29751
- [13] Elias Fernández Domingos, Francisco C. Santos, and Tom Lenaerts. 2023. EGT-tools: Evolutionary game dynamics in Python. *iScience* 26, 4 (2023), 106419. doi:10.1016/j.jsci.2023.106419
- [14] Elias Fernández Domingos, Inés Terrucha, Rémi Suchon, Jelena Grujić, Juan C. Burguillo, Francisco C. Santos, and Tom Lenaerts. 2022. Delegation to artificial agents fosters prosocial behaviors in the collective risk dilemma. *Scientific Reports* 12 (2022), 8492. doi:10.1038/s41598-022-11518-9
- [15] Apostolos Filippas, John J. Horton, and Benjamin S. Manning. 2024. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?. In *Proceedings of the 25th ACM Conference on Economics and Computation (EC ’24)*. ACM, New Haven, CT, USA, 614–615. doi:10.1145/3670865.3673513
- [16] Nicolò Fontana, Francesco Pierri, and Luca Maria Aiello. 2025. Nicer than Humans: How Do Large Language Models Behave in the Prisoner’s Dilemma? *Proceedings of the International AAAI Conference on Web and Social Media* 19 (2025), 522–535. doi:10.1609/icwsm.v19i1.35829
- [17] Drew Fudenberg and Lorens A. Imhof. 2006. Imitation processes with small mutations. *Journal of Economic Theory* 131, 1 (2006), 251–262. doi:10.1016/j.jet.2005.04.006
- [18] David Guzman Piedrahita, Yongjin Yang, Mrinmaya Sachan, Giorgia Ramponi, Bernhard Schölkopf, and Zhijing Jin. 2025. Corrupted by Reasoning: Reasoning Language Models Become Free-Riders in Public Goods Games. In *Conference on Language Modeling (COLM)*. OpenReview.net, Montreal, Canada, 37 pages. https://openreview.net/forum?id=kH6LOHGJel
- [19] Lewis Hammond, Alan Chan, Jesse Clifton, Jason Hoelscher-Obermaier, Akbir Khan, Euan McLean, Chandler Smith, Wolfram Barfuss, Jakob Foerster, Tomáš Gavenčák, The Anh Han, Edward Hughes, Vojtěch Kovářik, Jan Kulveit, Joel Z. Leibo, Caspar Oesterheld, Christian Schroeder de Witt, Nisarg Shah, Michael Wellman, Paolo Bova, Theodor Cîmpeanu, Carson Ezell, Quentin Feuillade-Montixi, Matija Franklin, Esben Kran, Igor Krawczuk, Max Lamparth, Niklas Lauffer, Alexander Meinke, Sumeet Motwani, Anka Reul, Vincent Conitzer, Michael Dennis, Iason Gabriel, Adam Gleave, Gillian Hadfield, Nika Haghtalab, Atoosa Kasirzadeh, Sébastien Krier, Kate Larson, Joel Lehman, David C. Parkes, Georgios Piliouras, and Iyad Rahwan. 2025. Multi-Agent Risks from Advanced AI. Cooperative AI Foundation, Technical Report #1. arXiv:2502.14143 doi:10.48550/arXiv.2502.14143
- [20] The Anh Han. 2022. Emergent behaviours in multi-agent systems with Evolutionary Game Theory. *AI Communications* 35, 4 (2022), 327–337. doi:10.3233/AIC-220104
- [21] The Anh Han, Manh Hong Duong, and Matjaž Perc. 2024. Evolutionary mechanisms that promote cooperation may not promote social welfare. *Journal of the Royal Society Interface* 21, 220 (2024), 20240547. doi:10.1098/rsif.2024.0547
- [22] The Anh Han, Luis Moniz Pereira, and Tom Lenaerts. 2017. Evolution of commitment and level of participation in public goods games. *Autonomous Agents and Multi-Agent Systems* 31, 3 (2017), 561–583.
- [23] The Anh Han, Luís Moniz Pereira, and Francisco C. Santos. 2012. The emergence of commitments and cooperation. In *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2012)*. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 559–566.
- [24] Jen-tse Huang, Eric John Li, Man Ho Lam, Tian Liang, Wenxuan Wang, Youliang Yuan, Wenxiang Jiao, Xing Wang, Zhaopeng Tu, and Michael R. Lyu. 2025. Competing Large Language Models in Multi-Agent Gaming Environments. In *International Conference on Learning Representations (ICLR)*. OpenReview.net, Singapore, 65639–65681. https://proceedings.iclr.cc/paper_files/paper/2025/hash/a46ad2bf0ca0e16ef8857e188991ad7-Abstract-Conference.html
- [25] Edward Hughes, Joel Z. Leibo, Matthew Phillips, Karl Tuyls, Edgar Dueñez-Guzman, Antonio Garcia Castañeda, Iain Dunning, Tina Zhu, Kevin McKee, Raphael Koster, Heather Roff, and Thore Graepel. 2018. Inequity Aversion Improves Cooperation in Intertemporal Social Dilemmas. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates, Inc., Red Hook, NY, USA, 3330–3340. https://proceedings.neurips.cc/paper_files/paper/2018/hash/7fe637fd6d02b8f0adf67dc36aed93-Abstract.html
- [26] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoi Tran-Le, Alessio Buscemi, Hong Trang Le, and The Anh Han. 2025. Understanding LLM Agent Behaviours via Game Theory: Strategy Recognition, Biases and Multi-Agent Dynamics. arXiv:2512.07462 doi:10.48550/arXiv.2512.07462
- [27] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoi Tran-Le, Alessio Buscemi, Hong Trang Le, and The Anh Han. 2026. Payoff scaling shapes cooperation in LLM agents across languages. arXiv:2601.19082 doi:10.48550/arXiv.2601.19082
- [28] Shivani Kumar, Adarsh Bharathwaj, and David Jurgens. 2026. Cooperative Profiles Predict Multi-Agent LLM Team Performance in AI for Science Workflows. arXiv:2604.20658 doi:10.48550/arXiv.2604.20658
- [29] Joel Z. Leibo, Edgar A. Dueñez-Guzman, Alexander Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Koster, Jayd Matyas, Charlie Beattie, Igor Mordatch, and Thore Graepel. 2021. Scalable Evaluation of Multi-Agent Reinforcement Learning with Melting Pot. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*. PMLR, Virtual Event, 6187–6199. https://proceedings.mlr.press/v139/leibo21a.html
- [30] Joel Z. Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. 2017. Multi-agent Reinforcement Learning in Sequential Social Dilemmas. In *Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2017)*. IFAAMAS, São Paulo, Brazil, 464–473. https://www.ifaamas.org/Proceedings/aamas2017/pdfs/p464.pdf
- [31] Nunzio Lorè and Babak Heydari. 2024. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports* 14 (2024), 18490. doi:10.1038/s41598-024-69032-z
- [32] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O. Jackson. 2024. A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences* 121, 9 (2024), e2313925121. doi:10.1073/pnas.2313925121
- [33] Ramona Merhej, Fernando P. Santos, Francisco S. Melo, and Francisco C. Santos. 2021. Cooperation between Independent Reinforcement Learners under Wealth Inequality and Collective Risks. In *Proceedings of the 20th International Conference on Autonomous Agents and MultiAgent Systems (Virtual Event, United Kingdom) (AAMAS ’21)*. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 898–906. https://www.ifaamas.org/Proceedings/aamas2021/pdfs/p898.pdf
- [34] Manfred Milinski, Ralf D. Sommerfeld, Hans-Jürgen Krambeck, Floyd A. Reed, and Jochem Marotzke. 2008. The collective-risk social dilemma and the prevention of simulated dangerous climate change. *Proceedings of the National Academy of Sciences* 105, 7 (2008), 2291–2294. doi:10.1073/pnas.0709546105

- [35] Shayegan Omidshafiei, Christos Papadimitriou, Georgios Piliouras, Karl Tuyls, Mark Rowland, Jean-Baptiste Lespiau, Wojciech M. Czarnecki, Marc Lanctot, Julien Perolat, and Remi Munos. 2019. α -Rank: Multi-Agent Evaluation by Evolution. *Scientific Reports* 9, 1 (2019), 9937. doi:10.1038/s41598-019-45619-9
- [36] Saptarshi Pal, Abhishek Mallela, Lenz Pracher, Chiyu Wei, Feng Fu, Santiago Schnell, and Martin A. Nowak. 2026. Large language models instantiate evolutionarily robust strategies of cooperation. *PNAS Nexus* 5, 6 (2026), pgag210. doi:10.1093/pnasnexus/pgag210
- [37] Thomas R. Palfrey and Howard Rosenthal. 1984. Participation and the provision of discrete public goods: a strategic analysis. *Journal of Public Economics* 24, 2 (1984), 171–193. doi:10.1016/0047-2727(84)90023-9
- [38] Kenneth Payne and Baptiste Alloui-Cros. 2025. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory. arXiv:2507.02618 doi:10.48550/arXiv.2507.02618
- [39] Phu Hoa Pham, Duy Minh Dao Sy, Trung Kiet Huynh, Phu Quy Nguyen Lam, Chi Nguyen Tran, Minh Trung Le, Phong Hao Le, Dinh Nam Nguyen, Thien Ky Nguyen Dong, Elias Fernandez Domingos, Hong Trang Le, and The Anh Han. 2026. Humans Are More Diverse: Frontier LLMs Show Extreme Policies in Idealised AI Development Races. arXiv:2608.01193 doi:10.48550/arXiv.2608.01193
- [40] Giorgio Piatti, Zhijing Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. 2024. Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., Vancouver, Canada, 111715–111759. doi:10.52202/079017-3548
- [41] Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W Crandall, Nicholas A Christakis, Iain D Couzin, Matthew O Jackson, et al. 2019. Machine behaviour. *Nature* 568, 7753 (2019), 477–486.
- [42] David G Rand, Corina E Tarnita, Hisashi Ohtsuki, and Martin A Nowak. 2013. Evolution of fairness in the one-shot anonymous ultimatum game. *Proceedings of the National Academy of Sciences* 110, 7 (2013), 2581–2586.
- [43] Francisco C. Santos and Jorge M. Pacheco. 2011. Risk of collective failure provides an escape from the tragedy of the commons. *Proceedings of the National Academy of Sciences* 108, 26 (2011), 10421–10425. doi:10.1073/pnas.1015648108
- [44] Fernando P Santos, Samuel Mascarenhas, Francisco C Santos, Filipa Correia, Samuel Gomes, and Ana Paiva. 2020. Picky losers and carefree winners prevail in collective risk dilemmas with partner selection. *Autonomous Agents and Multi-Agent Systems* 34, 2 (2020), 40.
- [45] Fernando P. Santos, Samuel F. Mascarenhas, Francisco C. Santos, Filipa Correia, Samuel Gomes, and Ana Paiva. 2019. Outcome-based Partner Selection in Collective Risk Dilemmas. In *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems* (Montreal, QC, Canada) (AAMAS '19). International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 1556–1564. <https://www.ifaamas.org/Proceedings/aamas2019/pdfs/p1556.pdf>
- [46] Mark E. Schaffer. 1988. Evolutionarily stable strategies for a finite population and a variable contest size. *Journal of Theoretical Biology* 132, 4 (1988), 469–478. doi:10.1016/S0022-5193(88)80085-7
- [47] Thomas Schmied, Jörg Bornschein, Jordi Grau-Moya, Markus Wulfmeier, and Razvan Pascanu. 2026. LLMs are Greedy Agents: Effects of RL Fine-tuning on Decision-Making Abilities. In *International Conference on Learning Representations (ICLR)*. OpenReview.net, Rio de Janeiro, Brazil, 127538–127571. https://proceedings.iclr.cc/paper_files/paper/2026/hash/cf2fda40de8520027af2eadcafdec6d-Abstract-Conference.html
- [48] Karl Sigmund, Hannelore De Silva, Arne Traulsen, and Christoph Hauert. 2010. Social learning promotes institutions for governing the commons. *Nature* 466, 7308 (2010), 861–863.
- [49] Oliver Slumbers, Joel Z. Leibo, and Marco A. Janssen. 2025. Using Large Language Models to Simulate Human Behavioural Experiments: Port of Mars. arXiv:2506.05555 doi:10.48550/arXiv.2506.05555
- [50] Peter Stone, Gal A. Kaminka, Sarit Kraus, and Jeffrey S. Rosenschein. 2010. Ad Hoc Autonomous Agent Teams: Collaboration without Pre-Coordination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 24. AAAI Press, Atlanta, Georgia, USA, 1504–1509. doi:10.1609/aaai.v24i1.7529
- [51] Alessandro Tavoni, Astrid Dannenberg, Giorgos Kallis, and Andreas Löschel. 2011. Inequality, communication, and the avoidance of disastrous climate change in a public goods game. *Proceedings of the National Academy of Sciences* 108, 29 (2011), 11825–11829. doi:10.1073/pnas.1102493108
- [52] Inês Terrucha, Elias Fernández Domingos, Francisco C. Santos, Pieter Simoons, and Tom Lenaerts. 2024. The art of compensation: How hybrid teams solve collective-risk dilemmas. *PLOS ONE* 19, 2 (2024), e0297213. doi:10.1371/journal.pone.0297213
- [53] Emanuel Tewolde, Xiao Zhang, David Guzman Piedrahita, Vincent Conitzer, and Zhijing Jin. 2026. CoopEval: Benchmarking Cooperation-Sustaining Mechanisms and LLM Agents in Social Dilemmas. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. PMLR, Seoul, South Korea, 65 pages. <https://icml.cc/virtual/2026/poster/66503>
- [54] Arne Traulsen and Martin A. Nowak. 2006. Evolution of cooperation by multilevel selection. *Proceedings of the National Academy of Sciences* 103, 29 (2006), 10952–10955. doi:10.1073/pnas.0602530103
- [55] Arne Traulsen, Martin A. Nowak, and Jorge M. Pacheco. 2006. Stochastic dynamics of invasion and fixation. *Physical Review E* 74, 1 (2006), 011909. doi:10.1103/PhysRevE.74.011909
- [56] Karl Tuyls, Julien Perolat, Marc Lanctot, Joel Z. Leibo, and Thore Graepel. 2018. A Generalised Method for Empirical Game Theoretic Analysis. In *Proceedings of the 17th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2018)*. IFAAMAS, Stockholm, Sweden, 77–85. <https://ifaamas.org/Proceedings/aamas2018/pdfs/p77.pdf>
- [57] Aron Vallinder and Edward Hughes. 2025. Cultural Evolution of Cooperation among LLM Agents. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*. IFAAMAS, Detroit, MI, USA, 2771–2773. <https://www.ifaamas.org/Proceedings/aamas2025/pdfs/p2771.pdf>
- [58] Michael P. Wellman. 2006. Methods for Empirical Game-Theoretic Analysis (Extended Abstract). In *Proceedings of the Twenty-First International Conference on Artificial Intelligence (AAAI-06)*. AAAI Press, Boston, Massachusetts, USA, 1552–1555. <https://aaai.org/Papers/AAAI/2006/AAAI06-248.pdf>
- [59] Richard Willis, Jianing Zhao, Yali Du, and Joel Z. Leibo. 2026. Evaluating Collective Behaviour of Hundreds of LLM Agents. Strategic Engineering Workshop on LLMs and Game Theory (SE@AAMAS 2026). arXiv:2602.16662 doi:10.48550/arXiv.2602.16662
- [60] Ioannis Zisis, Sibilla Di Guida, The Anh Han, Georg Kirchsteiger, and Tom Lenaerts. 2015. Generosity motivated by acceptance-evolutionary analysis of an anticipation game. *Scientific reports* 5, 1 (2015), 18076.